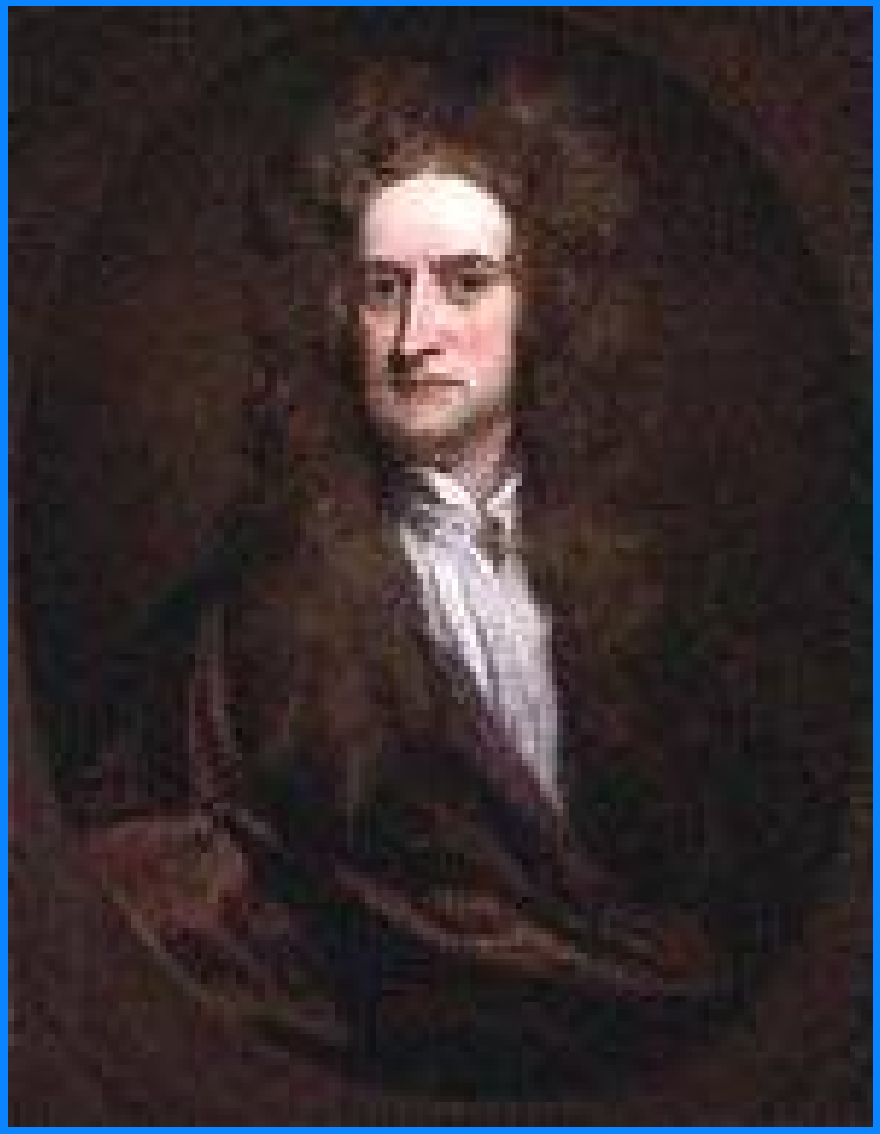
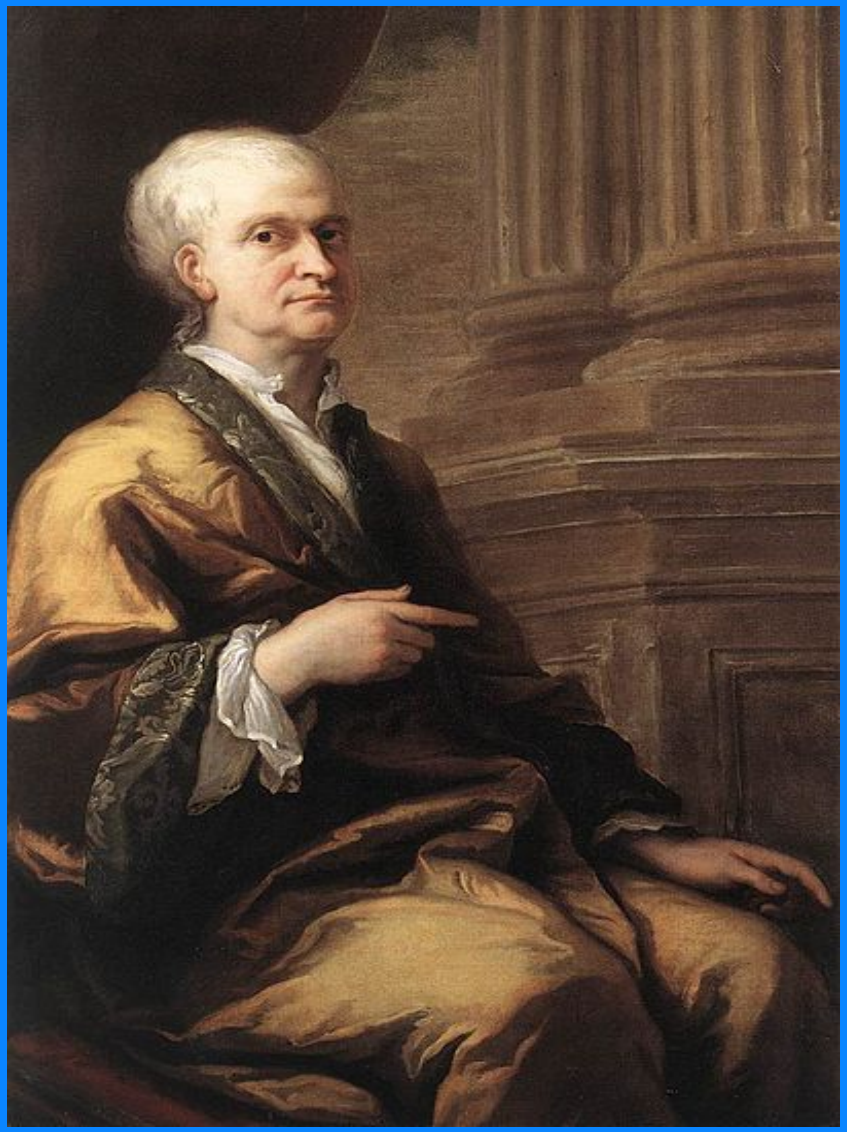
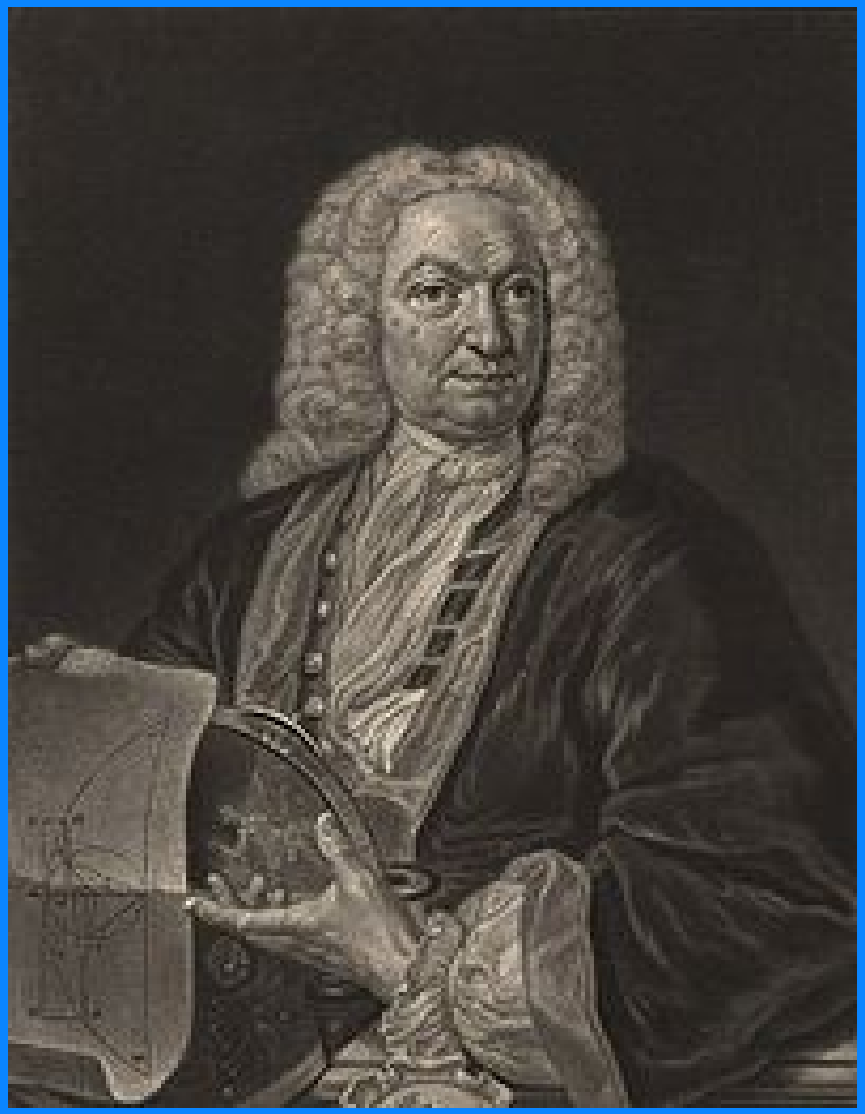
Introduction to Differential Equations





Sir Isaac Newton (1643-1727)



Johann Bernoulli (1667–1748)



Leonhard Paul Euler (1707–1783)

Basic Concepts

- ODE vs. PDE
- Dependent Variables vs. Independent Variables
- Order
- Linear vs. Nonlinear
- Solutions

Ordinary differential equations

Definition:

A differential equation is an equation containing an unknown function and its derivatives.

Examples:.

1. $$\frac{dy}{dx} = 2x + 3$$

2. $$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + ay = 0$$

3. $$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + 6y = 3$$

y is dependent variable and ***x*** is independent variable, and these are ordinary differential equations

Partial Differential Equation

Examples:

1. $$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$$

u is dependent variable and ***x and y*** are independent variables, and is partial differential equation.

2. $$\frac{\partial^4 u}{\partial x^4}+\frac{\partial^4 u}{\partial t^4}=0$$

3. $$\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial t^2}-\frac{\partial u}{\partial t}$$

u is dependent variable and ***x and t*** are independent variables

Order of Differential Equation

The order of the differential equation is order of the highest derivative in the differential equation.

Differential Equation	ORDER
$\frac{dy}{dx} = 2x + 3$	1
$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 9y = 0$	2
$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + 6y = 3$	3

Degree of Differential Equation

The degree of a differential equation is power of the highest order derivative term in the differential equation.

Differential Equation	Degree
$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + ay = 0$	1
$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + 6y = 3$	1
$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 + 3 = 0$	3

Linear Differential Equation

A differential equation is linear, if

1. **dependent variable and its derivatives are of degree one,**
2. **coefficients of a term does not depend upon dependent variable.**

Example: 1. $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 9y = 0.$

is linear.

Example: 2. $\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^4 + 6y = 3$

is non - linear because in 2nd term is not of degree one.

Example: 3.

$$x^2 \frac{d^2y}{dx^2} + y\frac{dy}{dx} = x^3$$

is non - linear because in 2nd term coefficient depends on *y*.

Example: 4.

$$\frac{dy}{dx} = \sin y$$

is non - linear because $\sin y = y - \frac{y^3}{3!} + -$ is non – linear

9. Table 1. Classify each differential equation

No	Differential Equations	Ordinary or Partial	Linear or nonlinear	Order	Degree	Independent variables	Dependent variables
1.	$y' = x + 6y$						
2.	$y'' = 4y + y^3$						
3.	$\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} - 2y = x^3$						
4.	$y'' + 2xy' + 4y = \cos 2x$						
5.	$\frac{dy}{dx} = \frac{x^2-1}{y+4}$						
6.	$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$						
7.	$\frac{\partial^2 u}{\partial x^2} = c^2 \frac{\partial^2 u}{\partial t^2}$						

It is Ordinary/partial Differential equation of order… and of degree…, it is linear / non linear, with independent variable…, and dependent variable….

1st – order Differential Equation

1. Derivative form:

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

2. Differential form:

$$(1+x)dy - ydx = 0$$

3. General form:

$$\frac{dy}{dx} = f(x, y) \qquad \text{or} \qquad f(x, y, \frac{dy}{dx}) = 0.$$

Second order Differential Equation

$$f(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}) = O.$$

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = g(x)$$

nth – order Linear Differential Equation

1. nth – order linear differential equation with constant coefficients.

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + ... + a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = g(x)$$

2. nth – order linear differential equation with variable coefficients

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + ... + a_2(x)\frac{d^2 y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

Why study differential equations?

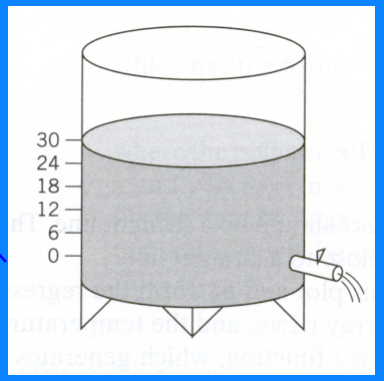
- Some examples of fields using differential equations in their analysis include:

  - Solid mechanics & motion
  - heat transfer & energy balances
  - vibrational dynamics & seismology
  - aerodynamics & fluid dynamics
  - electronics & circuit design
  - population dynamics & biological systems
  - climatology and environmental analysis
  - options trading & economics

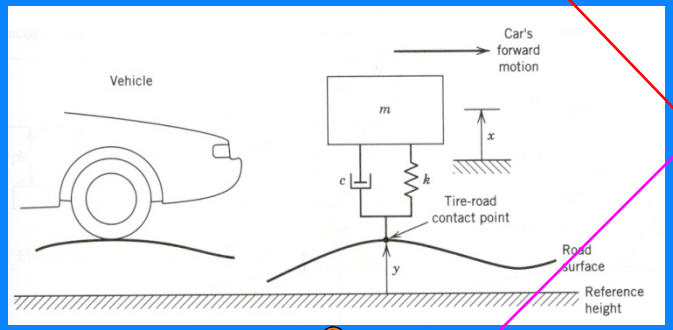
Examples of Fields Using Differential Equations in Their Analysis



$$A\frac{dh}{dt} = -C_d A_o \sqrt{2gh}$$

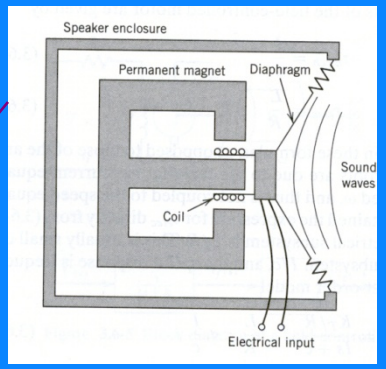


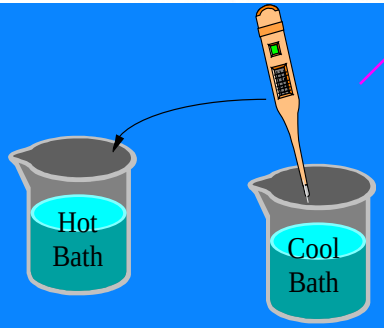
$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = c\frac{dy}{dt} + ky$$



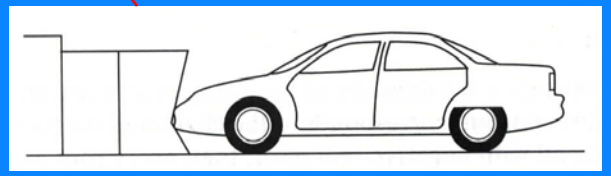
$$\frac{dH}{dt} = -k(H - 70)$$

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = 0$$

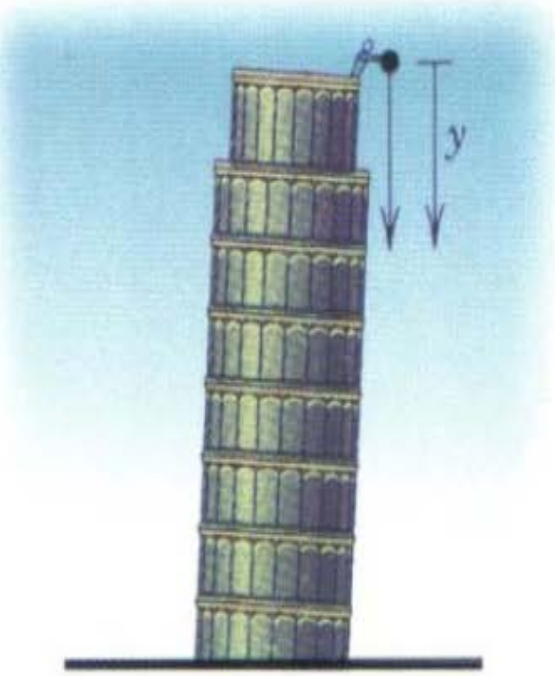




$$m\frac{d^2x}{dt^2} = -c\frac{dx}{dt} - kx + k_f i$$



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Some Applications of Differential Equations



การหล่นของก้อนหิน

$$y'' = g$$



การกระโดดร่ม

$$mv' = mg - bv^2$$

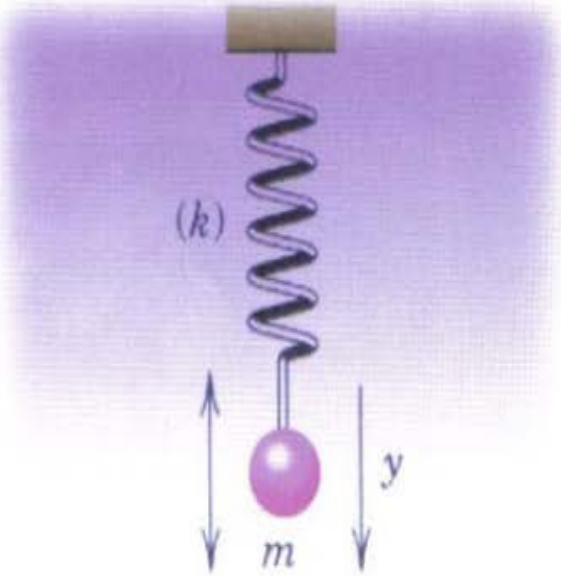


การไหลออกของน้ำ

$$h' = -k\sqrt{h}$$

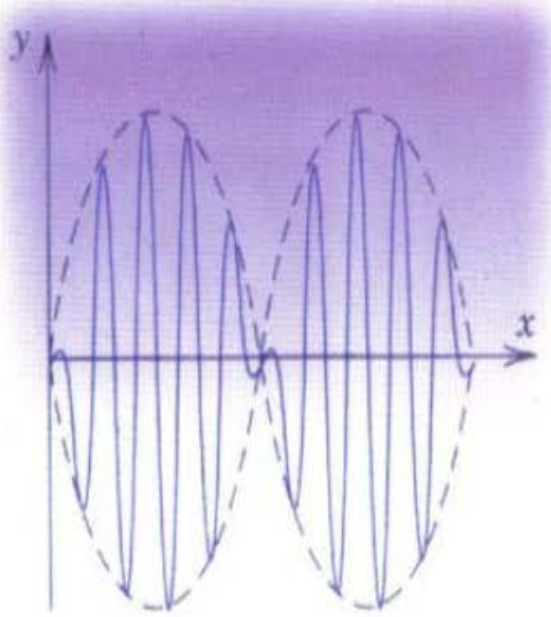
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Some Applications of Differential Equations



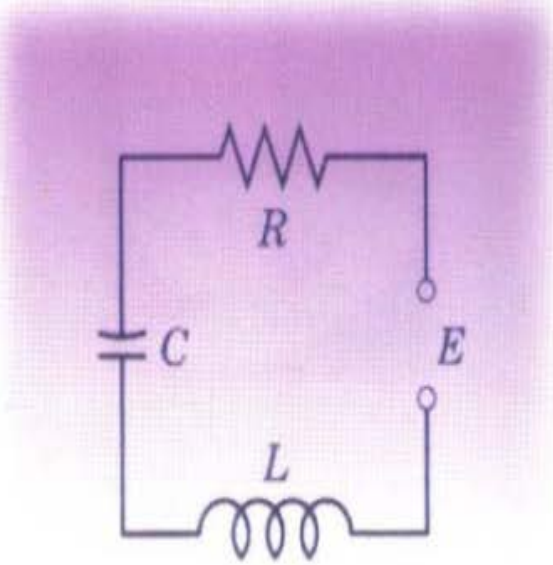
การสั่นของมวลบนสปริง

$$my'' + ky = 0$$



BEATS ของระบบของการสั่น

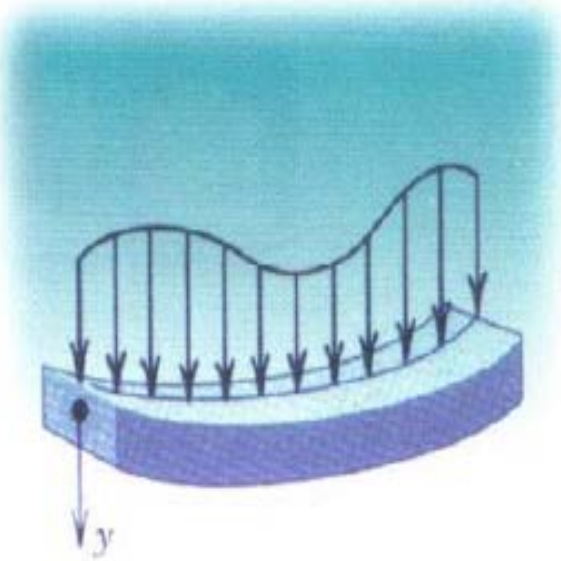
$$y'' + \omega_0^2 y = \cos\ \omega t, \omega_0 \approx \omega$$



กระแสในวงจร RLC

$$LI'' + RI' + \frac{1}{C}I = E'$$

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Some Applications of Differential Equations



คานที่ผิดปกติ

$$Ely^{IV} = f(x)$$



การเคลื่อนที่ของลูกตุ้ม

$$L'' + g \sin \theta = 0$$



โมเดลของการล่าเหยื่อ

$$y_1' = ay_1 - by_1y_2$$

$$y_2' = ky_1y_2 - ly_2$$

FIRST ORDER DIFFERENTIAL EQUATIONS

Basic Concepts

- The **first-order** differential equation contain only **y'** and may contain ***y*** and given function of ***x.***

$$F(x, y, y') = 0$$

or

$$y' = f(x, y) \qquad \text{——} (*)$$

A solution of a given **first-order** differential equation (*) on some open interval **a<x<b** is a function **y=h(x)** that has a derivative **y'=h(x)** and satisfies (*) for all x in that interval.

Basic Concepts

- ***Explicit Solution***

$$y = h(x)$$

- ***Implicit Solution***

$$H(x, y) = 0$$

Basic Concepts

- General solution *vs.* Particular solution

$$y' = \cos x$$
$$y = \sin x + c$$
$$c = -3, -2, \ldots$$

- General solution
  - arbitrary constant c
- Particular solution
  - choose a specific c

Basic Concept

- **Singular solutions**
- Def : A differential equation may sometimes have an additional solution that cannot be obtained from the **general solution** and is then called a singular solution.

- Example $y'^2 - xy' + y = 0$

The general solution : **$y=cx-c^2$**

A singular solution : **$y=x^2/4$**

C-discriminant

$$\begin{cases} \Phi(x, y, C) = 0 \\ \dfrac{\partial \Phi(x, y, C)}{\partial C} = 0 \end{cases}$$

$$y - cx + c^2 = 0$$

$$x - 2c = 0$$

Example 0. Consider the DE

$$y' = ky$$

- ***General Solution***

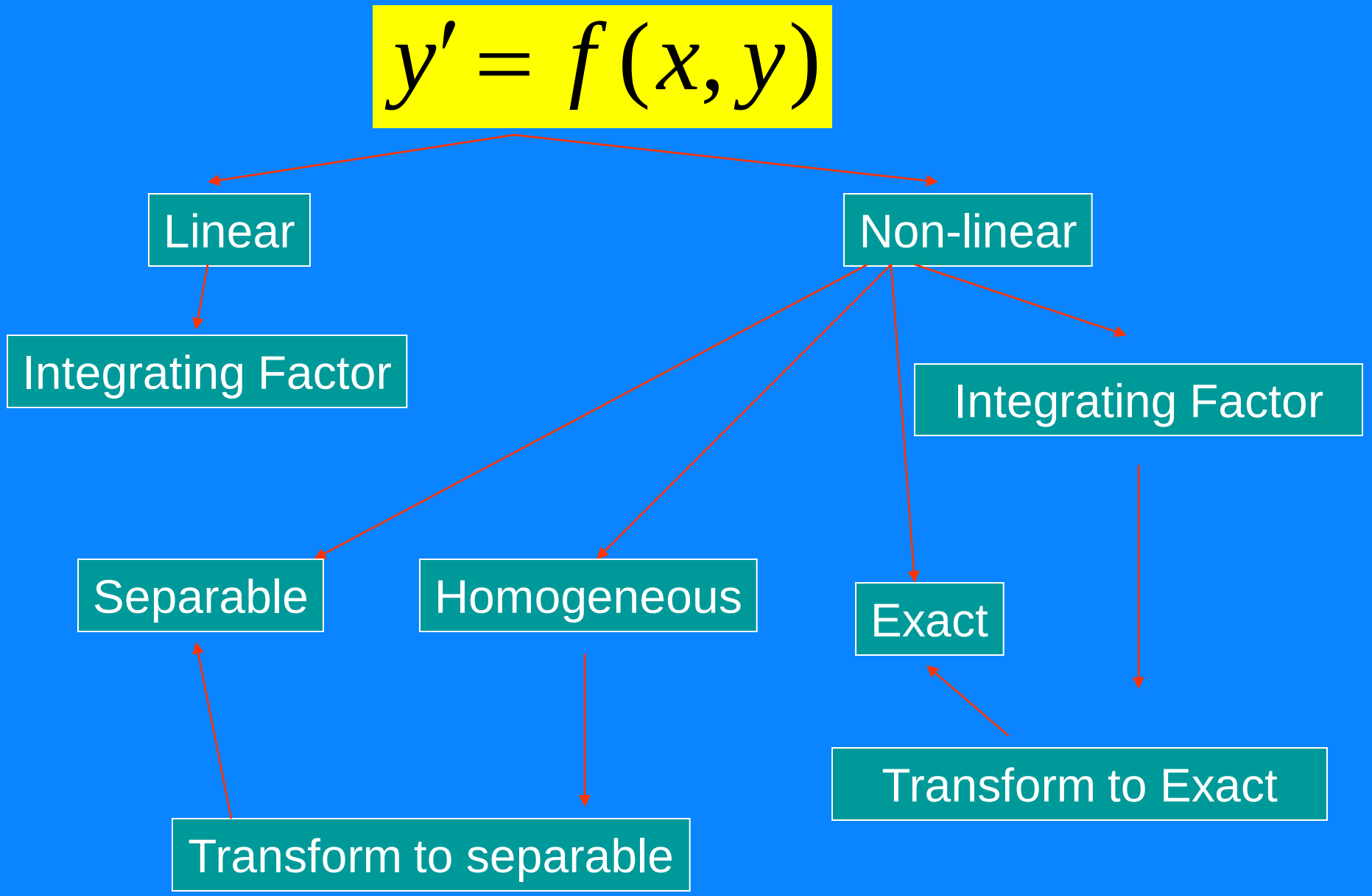
$$y(t) = ce^{kt}$$

- ***Particular Solution*** for y(0)=2 (***initial condition***)

$$y(t) = 2e^{kt}$$

Various methods of solving first order differential equations.

- Variables separable
- Homogeneous equations
- Exact equations
- Integrating factor
- Bernoulli's Equation


$y' = f(x, y)$
Linear
Non-linear
Integrating Factor
Integrating Factor
Separable
Homogeneous
Exact
Transform to Exact
Transform to separable

Variables Separable

The first-order differential equation

$$\frac{dy}{dx} = f(x, y) \qquad (1)$$

is called separable provided that $f(x, y)$ can be written as the product of a function of x and a function of y.

Suppose we can write the above equation as

$$\frac{dy}{dx} = g(x)h(y)$$

We then say we have " separated " the variables. By taking $h(y)$ to the LHS, the equation becomes

$$\frac{1}{h(y)}dy = g(x)dx$$

Integrating, we get the solution as

$$\int \frac{1}{h(y)}dy = \int g(x)\,dx + c$$

where c is an arbitrary constant.

Example 1. Consider the DE $\frac{dy}{dx} = y$

Separating the variables, we get

$$\frac{1}{y}dy = dx$$

Integrating we get the solution as

$$\ln|y| = x + c$$

or $y = ce^x$, C is an arbitrary constant.

Example 2. Consider the DE

$$3x^2 \ln y\, dx + \frac{x^3}{y} dy = 0$$

Separating the variables, we get

$$\frac{1}{y \ln y} dy = -\frac{3}{x} dx$$

Integrating we get the solution as

$$\ln|\ln y| = -3\ln|x| + c$$

or $\ln y = \frac{c}{x^3}$,C is an arbitrary constant.

Homogeneous equations

Definition A function $f(x, y)$ is said to be homogeneous of degree n in x, y if

$$f(tx, ty) = t^n f(x, y)$$ for all t, x, y

Examples $f(x, y) = x^2 - 2xy - y^2$

is homogeneous of degree 2.

$$f(x, y) = \frac{y}{x} + \sin\left(\frac{y - x}{x}\right)$$

is homogeneous of degree 0.

A first order DE $M(x,y)\,dx + N(x,y)\,dy = 0$

is called **homogeneous** if $M(x,y),\ N(x,y)$

are homogeneous functions of *x* and *y* of the **same degree**.

This DE can be written in the form $\dfrac{dy}{dx} = f(x,y)$

where $f(x,y) = -M(x,y)/N(x,y)$ is

clearly homogeneous of degree 0.

The substitution $y = z\,x$ converts the given equation into **"variables separable"** form and hence can be solved. (Note that z is also a (new) variable,) We illustrate by means of examples.

Example 3. Solve the DE

$$(3x^2 - y^2)\,dy - 2xy\,dx = 0$$

That is

$$\frac{dy}{dx} = \frac{2xy}{(3x^2 - y^2)}$$

Let $y = z\,x$. Hence we get

$$z + x\frac{dz}{dx} = \frac{2z\,x^2}{(3x^2 - z^2x^2)} = \frac{2z}{(3 - z^2)}$$

or $$x\frac{dz}{dx}=\frac{2z}{(3-z^2)}-z=\frac{-z+z^3}{(3-z^2)}$$

Separating the variables, we get

$$\frac{(3-z^2)}{z^3-z}dz=\frac{dx}{x}$$

Integrating we get

$$\int\frac{(3-z^2)}{z^3-z}dz=\int\frac{dx}{x}$$

We express the LHS integral by partial fractions.

$$\int\left(-\frac{3}{z}+\frac{1}{z+1}+\frac{1}{z-1}\right)dz=\int\frac{dx}{x}+\ln c$$

or $z^{-3}(z+1)(z-1)=cx$, ***c*** is an arbitrary constant.

Noting $z = y/x$, the solution is:

$(y+x)(y-x)=c\,y^3$, ***c*** is an arbitrary constant

or $x^2-y^2=c\,y^3$, ***c*** is an arbitrary constant

Working Rule to solve a HDE:

1. Put the given equation in the form

$$M(x,y)dx+N(x,y)dy=0$$

2. Check M and N are Homogeneous function of the same degree.

3. Let $y=zx$

4. Differentiate y = z x to get

$$\frac{dy}{dx} = z + x\frac{dz}{dx}$$

5. Solve the equation for z by separating the variables.

6. Replace z by y/x and simplify.

Example 4. Solve the DE

$$y' = \frac{y}{x} + \sin\left(\frac{y-x}{x}\right)$$

Let $y = z\,x$. Hence we get

$$z + x\frac{dz}{dx} = z + \sin(z-1) \quad \text{or} \quad x\frac{dz}{dx} = \sin(z-1)$$

Separating the variables, we get $\dfrac{1}{\sin(z-1)}\,dz = \dfrac{dx}{x}$

Integrating we get $\operatorname{cosec}(z\text{-}1) - \cot(z\text{-}1) = c\,x$

where $z = \dfrac{y}{x}$ and c is an arbitrary constant.

Non-homogeneous equations

We shall now see that some equations can be brought to homogeneous form by appropriate substitution.

Example 5. Solve the DE

$$(2x+y+1)\,dx+(x+3y+2)\,dy=0$$

That is

$$\frac{dy}{dx}=-\frac{(2x+y+1)}{(x+3y+2)}$$

We shall now put $x = u+h, y = v+k$

where h, k are constants (to be chosen).

Hence the given DE becomes

$$\frac{dv}{du} = -\frac{(2u+v+2h+k+1)}{(u+3v+h+3k+2)}$$

We now choose h, k such that

$$\begin{aligned} 2h+\ \ k+1&=0 \\ h+3k+2&=0 \end{aligned}$$

Hence $h = -\frac{1}{5}, k = -\frac{3}{5}$

Hence the DE becomes

$$\frac{dv}{du} = -\frac{(2u+v)}{(u+3v)}$$

which is homogeneous in u and v.

Let $v = z\,u$. Hence we get $$z + u\frac{dz}{du} = -\frac{(2+z)}{(1+3z)}$$

or $$u\frac{dz}{du} = -\frac{(2+z)}{(1+3z)} - z = \frac{(-2-2z-3z^2)}{(1+3z)}$$

Separating the variables, we get

$$\frac{(1+3z)}{2+2z+3z^2}dz = -\frac{du}{u}$$

Integrating we get $u^2(2+2z+3z^2) = c$

i.e. $2u^2 + 2uv + 3v^2 = c$

or $2(x+1/5)^2 + 2(x+\frac{1}{5})(y+\frac{3}{5}) + 3(y+\frac{3}{5})^2 = c$

Example 6. Solve the DE

$$(6x+4y+3)\,dx+(3x+2y+2)\,dy=0$$

That is $$\frac{dy}{dx}=-\frac{(6x+4y+3)}{(3x+2y+2)}$$

Now the previous method does not work as the lines

$$6x+4y+3=0$$
$$3x+2y+2=0$$

are parallel. We now put $u = 3x + 2y$.

The given DE becomes

$$\frac{1}{2}\left(\frac{du}{dx}-3\right)=-\frac{(2u+3)}{(u+2)}$$

or

$$\frac{du}{dx}=3-\frac{(4u+6)}{(u+2)}=\frac{-u}{u+2}$$

Separating the variables, we get

$$\frac{u+2}{u}du=-dx$$

Integrating, we get

$$u + \ln u^2 = -x + c$$

i.e.

$$3x + 2y + \ln(3x + 2y)^2 + x = c$$

EXACT DIFFERENTIAL EQUATIONS

A first order DE $M(x, y)\,dx + N(x, y)\,dy = 0$

is called an exact DE if there exits a function $f(x, y)$ such that

$$df = M(x, y)\,dx + N(x, y)\,dy$$

Here df is the 'total differential' of $f(x, y)$ and equals

$$\frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy$$

Hence the given DE becomes $df = 0$

Integrating, we get the solution as

$f(x, y) = c$, c an arbitrary constant

Thus the solution curves of the given DE are the 'level curves' of the function $f(x, y)$.

Example 7. The DE $y\,dx + x\,dy = 0$

is exact as it is $d\,(xy) = 0$

Hence the solution is: $x\,y = c$

Example 8. The DE $x\,dx + y\,dy = 0$

is exact as it is $d\,(x^2 + y^2) = 0$

Hence the solution is: $x^2 + y^2 = c$

Example 9. The DE $-\frac{1}{y}\sin(\frac{x}{y})dx + \frac{x}{y^2}\sin(\frac{x}{y})\,dy = 0$

is exact as it is $d(\cos(\frac{x}{y})) = 0$

Hence the solution is: $\cos(\frac{x}{y}) = c$

Test for exactness

Suppose $M(x, y)\,dx + N(x, y)\,dy = 0$

is exact. Hence there exists a function $f(x, y)$ such that

$$M(x, y)\,dx + N(x, y)\,dy = df = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy$$

Hence $$\frac{\partial f}{\partial x} = M, \quad \frac{\partial f}{\partial y} = N$$

Assuming all the 2nd order mixed derivatives of $f(x, y)$ are continuous, we get

$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial N}{\partial x}$$

Thus a necessary condition for exactness is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

We saw a necessary condition for exactness is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

We now show that the above condition is also sufficient for $M\,dx + N\,dy = 0$ to be exact. That is, if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

then there exists a function $f(x, y)$ such that

$$\frac{\partial f}{\partial x} = M, \quad \frac{\partial f}{\partial y} = N$$

Integrating $\dfrac{\partial f}{\partial x} = M$ partially w.r.t. x, we get

$$f(x, y) = \int_x M\, dx + g(y) \quad \ldots\ldots\ (*)$$

where $g(y)$ is a function of y alone

We know that for this $f(x, y)$, $\dfrac{\partial f}{\partial y} = N$

Differentiating (*) partially w.r.t. y, we get

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\int_x M\, dx + g'(y)$$ = N gives

or $$g'(y) = N - \frac{\partial}{\partial y}\int_x M\,dx \qquad \ldots (**)$$

We now show that the R.H.S. of (**) is independent of x and thus $g(y)$ (and so $f(x, y)$) can be found by integrating (**) w.r.t. y.

Prove the condition

$$\frac{\partial}{\partial x}\left(N - \frac{\partial}{\partial y}\int_x M\,dx \right) = \frac{\partial N}{\partial x} - \frac{\partial}{\partial x}\left(\frac{\partial}{\partial y}\int_x M\,dx \right)$$

$$= \frac{\partial N}{\partial x} - \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\int_x M\,dx \right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0$$

Note (1) The solution of the exact DE $d f = 0$ is $f(x, y) = c$.

Note (2) When the given DE is exact, the solution $f(x, y) = c$ is found as we did in the previous theorem. That is, we integrate M partially w.r.t. x to get

$$f(x, y) = \int_x M \, dx + g(y)$$

We now differentiate this partially w.r.t. y and equating to N, find $g'(y)$ and hence $g(y)$.

The following examples will help you in understanding this.

Example 10. Test whether the following DE is exact. If exact, solve it.

$$(x+\frac{2}{y})dy + y\,dx = 0$$

Here $M = y,\ N = (x+\frac{2}{y})$ $\quad \frac{\partial M}{\partial y} = 1 = \frac{\partial N}{\partial x}$

Hence exact.

Now $$f(x,y) = \int_x M\,dx = \int_x y\,dx = xy + g(y)$$

Differentiating partially w.r.t. y, we get

$$\frac{\partial f}{\partial y} = x + g'(y) = N = x + \frac{2}{y}$$

Hence $$g'(y) = \frac{2}{y}$$

Integrating, we get $$g(y) = 2\ln|y|$$

(Note that we have NOT put the arb constant)

Hence $$f(x,y) = xy + g(y) = xy + 2\ln|y|$$

Thus the solution of the given d.e. is

$$f(x,y) = c \quad \text{or} \quad xy + 2\ln|y| = c,$$ c an arb const.

Example 11. Test whether the following DE is exact. If exact, solve it.

$$(2xy^4 + \sin y)dx + (4x^2y^3 + x\cos y)dy = 0$$

Here $M = 2xy^4 + \sin y,\ N = 4x^2y^3 + x\cos y$

$$\frac{\partial M}{\partial y} = 8xy^3 + \cos y = \frac{\partial N}{\partial x}$$

Hence exact.

Now $$f(x,y) = \int_x M\,dx = \int_x (2xy^4 + \sin y)\,dx$$

$$= x^2y^4 + x\sin y + g(y)$$

Differentiating partially w.r.t. y, we get

$$\frac{\partial f}{\partial y} = 4x^2y^3 + x\cos y + g'(y)$$

$$= N = 4x^2y^3 + x\cos y \quad \text{Hence} \quad g'(y) = 0$$

Integrating, we get $g(y) = 0$

(Note that we have NOT put the arb constant) Hence

$$f(x,y) = x^2y^4 + x\sin y + g(y) = x^2y^4 + x\sin y$$

Thus the solution of the given d.e. is

$f(x,y) = c$ or $x^2y^4 + x\sin y = c,$ c an arb const.

In the above problems, we found $f(x, y)$ by integrating M partially w.r.t. x and then

equated $\dfrac{\partial f}{\partial y}$ *to* N.

We can reverse the roles of x and y. That is we can find $f(x, y)$ by integrating N partially

w.r.t. y and then equate $\dfrac{\partial f}{\partial x}$ *to* M.

The following problem illustrates this.

Example 12. Test whether the following DE is exact. If exact, solve it.

$$(1+y^2\sin 2x)dx-2y\cos^2 x\,dy=0$$

Here $M=1+y^2\sin 2x,$ $N=-2y\cos^2 x$

$$\frac{\partial M}{\partial y}=2y\sin 2x;\ \frac{\partial N}{\partial x}=4y\cos x\times\sin x=2y\sin 2x$$

Hence exact. Now

$$f(x,y)=\int_y N\,dy=\int_y -2y\cos^2 x\,dy$$

$$=-y^2\cos^2 x+g(x)$$

Differentiating partially w.r.t. x, we get

$$\frac{\partial f}{\partial x} = 2y^2 \cos x \times \sin x + g'(x) = y^2 \sin 2x + g'(x)$$

$$= M = 1 + y^2 \sin 2x \quad \text{gives} \quad g'(x) = 1$$

Integrating, we get $g(x) = x$

(Note that we have NOT put the arb constant) Hence

$$f(x, y) = -y^2 \cos^2 x + g(x) = -y^2 \cos^2 x + x$$

Thus the solution of the given d.e. is

$f(x, y) = c$ or $x - y^2 \cos^2 x = c,$ c an arb const.

Integrating Factors

The DE $(xy-1)\,dx+(x^2-xy)\,dy=0$ is NOT exact

but becomes exact when multiplied by $\frac{1}{x}$ as it becomes

$$\left(y-\frac{1}{x}\right)dx+(x-y)\,dy=0$$

i.e.

$$d\left(xy-\ln x-\frac{1}{2}y^2\right)=0$$

We say $\frac{1}{x}$ is an Integrating Factor of the given DE

Definition: If on multiplying by $\mu(x, y)$, the DE

$$M\,dx + N\,dy = 0$$

becomes an exact DE, we say that $\mu(x, y)$ is an Integrating Factor of the above DE

$$\frac{1}{xy}, \frac{1}{x^2}, \frac{1}{y^2}$$ are all integrating factors of

the non-exact DE $$y\,dx - x\,dy = 0$$

We give some methods of finding integrating factors of an non-exact DE

Problem. Under what conditions will the DE

$$M\,dx + N\,dy = 0$$

have an integrating factor that is a function of x alone ?

Solution. Suppose $\mu = h(x)$ is an I.F.

Multiplying by $h(x)$ the above d.e. becomes

$$h(x)M\,dx + h(x)N\,dy = 0 \;\ldots\ldots(*)$$

Since (*) is an exact DE, we have

$$\frac{\partial}{\partial y}(h(x)M)=\frac{\partial}{\partial x}(h(x)N)$$

i.e. $$h(x)\frac{\partial M}{\partial y}+M\frac{\partial}{\partial y}h(x)=h(x)\frac{\partial N}{\partial x}+N\frac{\partial}{\partial x}h(x)$$

or $$h(x)\frac{\partial M}{\partial y}+M\times 0=h(x)\frac{\partial N}{\partial x}+N\,h'(x)$$

or $$h(x)(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x})=N\,h'(x)$$

or $$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=g(x)=\frac{h'(x)}{h(x)}$$

Hence if

$$\frac{h'(x)}{h(x)} = g(x) \quad \Rightarrow \quad \ln h(x) = \int g(x)dx$$

is a function of x alone, then $\mu = h(x) = e^{\int g(x)\,dx}$

is an integrating factor of the given DE

$$M\,dx + N\,dy = 0$$

Consider the DE $M\,dx + N\,dy = 0$

If $$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = g(y) = \frac{h'(y)}{h(y)}$$

is a function of *y* alone, then

$$\mu = h(y) = e^{\int g(y)\,dy}$$

is an integrating factor of the given DE

Example 13. Find an I.F. for the following DE and hence solve it.

Here $(x+3y^2)\,dx+2xy\,dy=0$

$M=(x+3y^2);\ N=2xy$

$$\frac{\partial M}{\partial y}=6y\neq 2y=\frac{\partial N}{\partial x}$$

Hence the given DE is not exact.

Now $$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N}=\frac{6y-2y}{2xy}=\frac{2}{x}=g(x),$$

a function of x alone. Hence

$$\mu=e^{\int g(x)\,dx}=e^{\int \frac{2}{x}dx}=x^2$$

is an integrating factor of the given DE

Multiplying by x^2, the given DE becomes

$$(x^3 + 3x^2y^2)\,dx + 2x^3y\,dy = 0$$

which is of the form $M\,dx + N\,dy = 0$

Note that now $M = (x^3 + 3x^2y^2);\ N = 2x^3y$

Integrating, we easily see that the solution is

$$\frac{x^4}{4} + x^3y^2 = c,$$

c an arbitrary constant.

Example 14. Find an I.F. for the following DE and hence solve it.

$$e^x\,dx+(e^x\cot y+2y\,\csc y)\,dy=0$$

Here $M=e^x;\ N=e^x\cot y+2y\,\csc y$

$$\frac{\partial M}{\partial y}=0\neq e^x\cot y=\frac{\partial N}{\partial x}$$

Hence the given DE is not exact.

Now $$\frac{\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}}{M}=\frac{e^{x}\cot y}{e^{x}}=\cot y=h(y),$$

a function of y alone. Hence

$$\mu=e^{\int h(y)\,dy}=e^{\int \cot y\,dy}=\sin y$$

is an integrating factor of the given DE

Multiplying by sin y, the given DE becomes

$$e^x \sin y\, dx + (e^x \cos y + 2y)\, dy = 0$$

which is of the form $M\, dx + N\, dy = 0$

Note that now

$$M = e^x \sin y;\ N = e^x \cos y + 2y$$

Integrating, we easily see that the solution is

$$e^x \sin y + y^2 = c,$$

c an arbitrary constant.

Problem. Under what conditions will the DE

$$M\,dx + N\,dy = 0$$

have an integrating factor that is a function of the product $z = x\,y$?

Solution. Suppose $\mu = h(z)$ is an I.F.

Multiplying by $h(z)$ the above d.e. becomes

$$h(z)M\,dx + h(z)N\,dy = 0 \;\ldots\ldots(*)$$

Since (*) is an exact DE, we have

$$\frac{\partial}{\partial y}(h(z)M)=\frac{\partial}{\partial x}(h(z)N)$$

i.e. $$h(z)\frac{\partial M}{\partial y}+M\frac{\partial}{\partial y}h(z)=h(z)\frac{\partial N}{\partial x}+N\frac{\partial}{\partial x}h(z)$$

or $$h(z)\frac{\partial M}{\partial y}+M\,h'(z)\times x=h(z)\frac{\partial N}{\partial x}+N\,h'(z)\times y$$

or $$h(z)(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x})=h'(z)(Ny-Mx)$$

or $$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{Ny-Mx}=\frac{h'(z)}{h(z)}$$

Hence if $$\frac{\dfrac{\partial M}{\partial y}-\dfrac{\partial N}{\partial x}}{Ny-Mx}=g(z)=\frac{h'(z)}{h(z)}$$

is a function of $z = x\,y$, then

$$\mu = h(z) = e^{\int g(z)\,dz}$$

is an integrating factor of the given DE

$$M\,dx + N\,dy = 0$$

Example 15. Find an I.F. for the following DE and hence solve it.

$$y\,dx+(x-2x^2y^3)\,dy=0$$

Here $$M=y;\ N=x-2x^2y^3$$

$$\frac{\partial M}{\partial y}=1\neq 1-4xy^3=\frac{\partial N}{\partial x}$$

Hence the given DE is not exact.

Now $$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{Ny-Mx}=\frac{1-(1-4xy^3)}{(xy-2x^2y^4)-xy}$$

$$=-\frac{2}{xy}=-\frac{2}{z}=g(z)$$

a function of $z=xy$. Hence

$$\mu=e^{\int g(z)\,dz}=e^{\int -\frac{2}{z}dz}=\frac{1}{z^2}=\frac{1}{x^2y^2}$$

is an integrating factor of the given DE

Multiplying by $\frac{1}{x^2y^2}$, the given DE becomes

$$\frac{1}{x^2y}dx+\left(\frac{1}{xy^2}-2y\right)dy=0$$

which is of the form $M\,dx+N\,dy=0$

Integrating, we easily see that the solution is

$$\frac{1}{xy}+y^2=c,$$

c an arbitrary constant.

Problem. Under what conditions will the DE

$$M\,dx + N\,dy = 0$$

have an integrating factor that is a function of the sum $z = x + y$?

Solution. Suppose $\mu = h(z)$ is an I.F.

Multiplying by $h(z)$ the above DE becomes

$$h(z)M\,dx + h(z)N\,dy = 0 \ldots\ldots(*)$$

Since (*) is an exact DE, we have

$$\frac{\partial}{\partial y}(h(z)M)=\frac{\partial}{\partial x}(h(z)N)$$

i.e. $$h(z)\frac{\partial M}{\partial y}+M\frac{\partial}{\partial y}h(z)=h(z)\frac{\partial N}{\partial x}+N\frac{\partial}{\partial x}h(z)$$

or $$h(z)\frac{\partial M}{\partial y}+M\,h'(z)=h(z)\frac{\partial N}{\partial x}+N\,h'(z)$$

or $$h(z)(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x})=h'(z)(N-M)$$

or $$\frac{\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}}{N-M}=\frac{h'(z)}{h(z)}$$

Hence if

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N - M} = g(z)$$

is a function of $z = x + y$, then

$$\mu = e^{\int g(z)\,dz}$$

is an integrating factor of the given DE

$$M\,dx + N\,dy = 0$$

Linear Equations

A linear first order equation is an equation that can be expressed in the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = b(x), \qquad (1)$$

where $a_1(x)$, $a_0(x)$, and $b(x)$ depend only on the independent variable x , not on y.

We assume that the function $a_1(x)$, $a_0(x)$, and $b(x)$ are continuous on an interval and that $a_1(x) \neq 0$ on that interval. Then, on dividing by $a_1(x)$, we can rewrite equation (1) in the standard form

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (2)$$

where P(x), Q(x) are continuous functions on the interval.

Let's express equation (2) in the differential form

$$[P(x)y - Q(x)]dx + dy = 0 \quad (3)$$

If we test this equation for exactness, we find

$$\frac{\partial M}{\partial y} = P(x) \quad and \quad \frac{\partial N}{\partial x} = 0$$

Consequently, equation (3) is exact only when P(x) = 0. It turns out that an integrating factor μ, which depends only on x, can easily obtained the general solution of (3).

Multiply (3) by a function μ(x) and try to determine μ(x) so that the resulting equation

$$[\mu(x)P(x)y - \mu(x)Q(x)]dx + \mu(x)dy = 0 \quad (4)$$

is exact. $$\frac{\partial M}{\partial y} = \mu(x)P(x) \quad and \quad \frac{\partial N}{\partial x} = \frac{d\mu(x)}{dx}$$

We see that (4) is exact if μ satisfies the DE

$$\mu(x)P(x) = \frac{d\mu(x)}{dx} \quad (5)$$

$$\mu(x) = \exp\left(\int P(x)dx\right) \quad (6)$$

Which is our desired IF

In (2), we multiply by μ(x) defined in (6) to obtain

$$\mu(x)\frac{dy}{dx}+\mu(x)P(x)y=\mu(x)Q(x) \qquad (7)$$

We know from (5) $\mu(x)P(x)=\frac{d\mu(x)}{dx}$

and so (7) can be written in the form

$$\underbrace{\mu(x)\frac{dy}{dx}+\frac{d\mu}{dx}(x)y}_{\frac{d}{dx}(\mu(x)y)}=\mu(x)Q(x)$$

$$\frac{d}{dx}(\mu(x)y) = \mu(x)Q(x) \quad (8)$$

Integrating (8) w.r.t. x gives

$$\mu(x)y = \int \mu(x)Q(x)dx + C$$

and solving for y yields

$$y = \mu(x)^{-1}\left(\int \mu(x)Q(x)dx + C\right)$$

Working Rule to solve a LDE:

1. Write the equation in the standard form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

2. Calculate the IF μ(x) by the formula

$$\mu(x) = \exp\left(\int P(x)dx\right)$$

3. Multiply the equation in standard form by μ(x) and recalling that the LHS is just

$$\frac{d}{dx}(\mu(x)y),$$ obtain

$$\frac{d}{dx}(\mu(x)y) = \mu(x)Q(x)$$

4. Integrate the last equation and solve for y by dividing by μ(x).

Example 16. Solve

$$x\cos x\left(\frac{dy}{dx}\right)+y\left(x\sin x+\cos x\right)=1$$

Solution :- Dividing by x cos x, throughout, we get

$$\frac{dy}{dx}+\left(\tan x+\frac{1}{x}\right)y=\frac{\sec x}{x}$$

$$\mu(x)=e^{\int P(x)dx}=e^{\int\left(\tan x+\frac{1}{x}\right)dx}$$

$$\mu(x) = e^{\ln|\sec x| + \ln|x|} = e^{\ln|\sec x|}\, e^{\ln|x|} = x\sec x$$

Multiply by $x\sec x$ yields

$$x\sec x\frac{dy}{dx} + (x\sec x\tan x + \sec x)y = \frac{\sec x}{x}x\sec x$$

$$\frac{d}{dx}(y\,x\sec x) = \frac{\sec x}{x}x\sec x$$

Integrate both side we get

$$yx\sec x = \int \sec^2 x\,dx + C = \tan x + C$$

Problem: Find the general solution of the equation

$$y - x + xy\cot x + xy' = 0.$$

Answer

$$xy\sin x = \sin x - x\cos x + C.$$

The usual notation $\frac{dy}{dx}$ implies that x is independent variable & y, the dependent variable. Sometimes it is helpful to replace x by y and y by x & work on the resulting equation.

* When diff equation is of the form

$$\frac{dx}{dy} + P(y).x = Q(y)$$

$$IF = e^{\int P(y)dy} \quad \& \quad solution \quad is \quad x \times IF = \int Q \times IF \, dy + c$$

Solve $y - xy' = y' y^2 e^y$

$$\Rightarrow y = \left(x + y^2 e^y\right)\frac{dy}{dx}$$

$$\Rightarrow y\frac{dx}{dy} = x + y^2 e^y$$

$$\frac{dx}{dy} - \frac{1}{y}x = y e^y$$

$$IF = e^{-\int \frac{1}{y} dy} = e^{-\ln y} = \frac{1}{y}$$

$$\frac{1}{y}\frac{dx}{dy} - \frac{1}{y^2}x = e^y$$

$$\therefore \frac{d}{dy}\left(\frac{x}{y}\right) = e^y$$

$$\Rightarrow \frac{x}{y} = e^y + c$$

$$\Rightarrow x = y e^y + cy$$

Bernoulli's Equation

- Def: ***Bernoulli*** equations

$$y' + P(x)y = Q(x)y^n$$

  - If n = 0, Bernoulli Eq. => First Order Linear Eq.
  - If n ≠ 0, let w = y^{1-n}

$$w' + (1-n)Pw = (1-n)Q$$

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \neq 0, n \neq 1$$

Dividing both sides of the DE by y^n, we obtained

$$y^{-n}\frac{dy}{dx} + P(x)y^{1-n} = Q(x)$$

Substitute

$$w = y^{1-n}$$

$$\frac{dw}{dx} = (1-n)y^{-n}\frac{dy}{dx}.$$

We obtained

$$\frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x),$$

A first order linear Differential Equation

Example 17. Solve the Bernoulli's equation

$$\frac{dy}{dx} + y = xy^3$$

Solution: Comparing the given DE with Standard form of the Bernoulli's equation

n=3 ➡

$$w = y^{1-3} = y^{-2}$$

$$\frac{dw}{dx} = (1-3)y^{-3}\frac{dy}{dx} = -2y^{-3}\frac{dy}{dx}.$$

Divinding given DE with y^{-3}

$$y^{-3}\frac{dy}{dx} + y^{-2} = x$$

$$-\frac{1}{2}\frac{dw}{dx} + w = x$$

$$\frac{dw}{dx} - 2w = -2x$$ is First order linear DE

Integrating Factor $\mu(x) = e^{\int (-2)dx} = e^{-2x}$

$$e^{-2x}\frac{dw}{dx} - 2e^{-2x}w = -2xe^{-2x}$$

$$\frac{d}{dx}\left(e^{-2x}w\right) = -2xe^{-2x}$$

$$e^{-2x}w = -2\int xe^{-2x}dx$$

$$e^{-2x}w = -2\left[-\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}\right] + c$$

$$e^{-2x}w = xe^{-2x} + \frac{1}{2}e^{-2x} + c$$

$$w = x + \frac{1}{2} + ce^{2x}$$

$$y^{-2} = x + \frac{1}{2} + ce^{2x}$$

is general solution

Summary

Separable	$g(y)dy = f(x)dx$
Homogeneous	$f(x,y) = -\frac{M(x,y)}{N(x,y)}$
Substitution	$g(u)du = f(x)dx$
Exact	$M(x,y)dx + N(x,y)dy = 0$
Integrating Factor	$\mu Pdx + \mu Qdy = 0$
Linear	$y' + p(x)y = r(x)$
Bernoulli	$y' + p(x)y = g(x)y^a$